Definitional Instability in Kinase Inhibitor Selectivity: An Empirical Characterization and Motivated Properties for a Well-Formed Measure

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Abstract

Kinase inhibitor selectivity is quantified using several competing definitions, including the S-score, selectivity entropy, Gini coefficient, and ratio-based measures, which are often used interchangeably. On four profiling datasets spanning three assay technologies, each curated to one reference target per human protein kinase, we show that their rankings depend on choices usually left implicit. Moving the activity boundary between the assay floor and the activity threshold reverses 8–13% of compound pairs under the S-score, entropy and Gini on three datasets, and over 20% on the fourth. The ratio depends strongly on which off-target it references. With entropy and Gini on a common log-affinity scale, the S-score, entropy and Gini agree more with one another than with the ratio among compounds with an active kinase. This grouping fails for the entropy as originally published. We state three properties for a selectivity measure: a reliability threshold, consistency across placements of the activity boundary, and monotonicity when a weakly bound kinase is added. No measure we assess satisfies the second, so the boundary must be fixed in the analysis protocol. Monotonicity holds for simple

measures such as a count of active kinases, but not for the Gini or the entropy with a low baseline. A gated, smooth-hinge entropy is monotone, but ranks compounds almost identically to the share of targets with detected binding. On small random panels, measures that follow this share recover rankings from fewer targets than the ratio and the published entropy. Recovery is limited mainly by the share of compounds that can still be scored.

Keywords: binding selectivity, cancer drug discovery, cheminformatics, kinase inhibition, research methodology.

Introduction

Kinase inhibitors are among the most intensely investigated compound classes in drug discovery, with 94 FDA-approved agents and several hundred in clinical development.[1] Because the human kinome is comprised of more than 500 proteins sharing a highly conserved ATP-binding site,[2] the design of inhibitors with controlled target profiles remains a central challenge. Selectivity is a primary criterion in lead optimization and a key determinant in the drug approval process.[3]

Despite its importance, no single consistent definition of selectivity exists. The field employs at least four distinct quantitative measures: the S-score,[4] selectivity entropy,[5] Gini coefficient,[6] and ratio-based measures, which compare the primary target with an off-target. These definitions are often used interchangeably, and their parameters (the threshold in the S-score and the reference off-target in ratio-based measures) are usually set by convention. The published entropy and Gini have no such parameter, but computing them requires choosing how affinities are weighted and how pairs without a measured affinity are treated. Standardizing them would remove one source of disagreement between laboratories and platforms.

Bosc et al.[7] showed that these definitions can produce substantially different compound

rankings on the same dataset, and traced individual disagreements to profile shape, such as a broad binder with one dominant target. The KInhibition platform[8] explicitly acknowledged that conflicting results across metrics are common. To our knowledge, which binding profile properties determine when the definitions agree or conflict has not been characterized systematically. Go/no-go decisions in lead optimization are often made on selectivity assessments whose sensitivity to the choice of definition is unknown, and inconsistent definitions make compounds hard to compare across studies.

In information theory, Shannon[9] settled which function should measure entropy by writing down a few properties a sensible measure must have and showing that only one formula satisfies them. We follow this precedent in part: we characterize how the competing measures behave, state properties a good measure should have, and examine which formulas satisfy them. We use four profiling datasets that span three assay technologies, each curated to one reference target per human protein kinase. Our contributions are as follows:

(i) we show that, with entropy and Gini on a common log-affinity scale, the S-score, entropy and Gini (the distribution-based family) agree more with each other than with the fold-selectivity ratio $R(k)$. This holds clearly on the two pK_d datasets among compounds with an active kinase, and with qualifications on the other two. With the entropy in its original association-constant form, this grouping fails on all three affinity datasets, by an amount that depends on how pairs without detected binding are treated;

(ii) we identify three sources of rank instability: the convention each definition uses to score compounds with no binding above the activity threshold, near-tied top targets, and broad profiles, often with one dominant target;

(iii) we state three properties, (D1) a reliability threshold, (D2) distributional consistency and (D3) monotonicity under weak off-target addition. No measure we assess satisfies D2: moving the activity boundary between the assay floor and the activity threshold

reverses 8–13% of compound pairs under the S-score, entropy and Gini on three datasets. D3 separates the measures;

(iv) we give a gated, smooth-hinge entropy that provably satisfies D3 and weights binding below the activity threshold continuously. On these data it ranks compounds almost identically to the entropy and to the share of targets with detected binding, which also satisfies D3;

(v) we evaluate recovery from random subpanels separately from the properties. Measures that follow the share of targets with detected binding recover rankings from fewer targets than measures that rest on the strongest targets, such as the ratio and the published entropy. On small panels, recovery is limited mainly by the share of compounds that can be scored.

Background

Existing Selectivity Metrics and Their Comparison

Several metrics have been proposed to quantify kinase inhibitor selectivity from profiling panel data. Karaman et al.[4] introduced the selectivity score (S-score), defined as the fraction of tested kinases bound with a dissociation constant below a fixed cutoff. Uitdehaag and Zaman[5] proposed the selectivity entropy, which treats the normalized binding profile as a probability distribution and computes its Shannon entropy. Graczyk[6] adapted the Gini coefficient from economics, treating the distribution of a compound's activity across the panel as an inequality distribution. Klaeger et al.[10] introduced the concentration- and target-dependent selectivity (CATDS) score, which quantifies the reduction in binding to a given target relative to the summed reduction across all detected targets at a specified inhibitor concentration.

Bosc et al.[7] subsequently proposed two further metrics, the window score and ranking score, and compared them against four previously published metrics (the S-score, Gini

coefficient, selectivity entropy, and a partition index) on the Davis and Anastassiadis datasets. They showed that compound rankings differ substantially depending on the metric applied, computed several metrics at more than one parameter value (S-score thresholds, window widths, ranking positions), and left a compound unranked under the S-score when no kinase cleared its threshold. Their window and ranking scores, which start from each compound's best affinity, were computed for every compound. They also traced individual disagreements to profile shape, such as a broad binder with one dominant target.

To our knowledge, attempts to reconcile the definitions have been limited. Miljkovic and Bajorath[11] compared CATDS-based selectivity assessments from Klaeger et al. with activity data mined from ChEMBL for the same compounds, finding broad consistency between the two sources. The KInhibition platform[8] addressed conflicting results across metrics by introducing its own composite score (combining on-target inhibition with an off-target inhibition penalty). More recently, the same group built a KIRHub resource[12] on a new profiling campaign of 86 approved kinase inhibitors against 758 kinases and disease-associated mutant variants. This resource offers users a choice between a further named metric (the KISS score, an on-target/off-target balance) and a separate Gini-based score.

To our knowledge, how much each parameter and implementation choice moves the rankings across datasets and assay technologies, and which profile properties determine when the definitions agree, have been illustrated for individual compounds but not characterized systematically. The present work addresses both.

Axiomatic Approaches to Ranking and Measurement

Precedent for an axiomatic characterization of measurement functions is established. Shannon[9] demonstrated that the entropy of a probability distribution is uniquely determined, up to a multiplicative constant, by three requirements: continuity in the probabilities, monotonic growth with the number of equally likely outcomes, and consistency under grouping of outcomes. Khinchin[13] subsequently gave an alternative axiomatization (continuity, maximality

on the uniform distribution, expansibility, and additivity over independent subsystems). The selectivity entropy of Uitdehaag and Zaman applies the Shannon formula to normalized binding profiles but provides no analogous axiomatic justification. The axioms that single out Shannon entropy, such as maximality on the uniform distribution and additivity over independent subsystems, characterize a measure of uncertainty, and none of them was chosen with selectivity in mind.

Axiomatic analysis can also show that no function meets a set of requirements. Arrow[14] proved this for social welfare functions, and Singleton and Booth[15] for a subset of candidate axioms for truth discovery, the problem of identifying true facts from conflicting reports. Of the three properties we state for selectivity, no measure we examine meets the second, and several simple constructions meet the other two (Results).

To our knowledge, this axiomatic style of analysis has not previously been applied to any selectivity or promiscuity measure in drug discovery. The present work proposes empirically motivated properties as a foundation for such an analysis.

Methods

Datasets

The Davis dataset[16] comprises binding affinities (K_d) for 68 small-molecule kinase inhibitors measured against 433 recombinant kinase constructs in a competition-binding assay. Affinity values are reported as dissociation constants and were converted to $\mathrm{p}K_d = -\log_{10}(K_d/10^9)$ prior to analysis, following the convention of the DeepDTA benchmark.[17] The released affinity matrix is complete (68×433, 29,444 entries), with all non-binding pairs assigned the floor value $\mathrm{p}K_d = 5.0$ (corresponding to a K_d of 10 μM). This follows DeepDTA,[17] and Bosc et al. used nearly the same value (10.1 μM).[7] It places non-binding pairs at the floor, above their true, unmeasured affinity.

The released constructs are not one per kinase. They include mutant, phosphorylation-

state and single-domain constructs of the same gene (for example, 12 EGFR constructs, and BRAF and BRAF(V600E)), 20 lipid-kinase constructs and three non-human kinases. For 26 of the 68 compounds, the two best-bound constructs are forms of the same gene. Selectivity across kinases needs one entry per kinase, chosen without reference to any compound's affinities. We therefore use one reference construct per human protein kinase gene: the wild-type construct, or its catalytic or N-terminal kinase domain where several domains are profiled. Genes profiled only as mutants are removed. This leaves 363 targets. Results on the released constructs, and with the best affinity per gene in place of the reference construct, are in Supporting Information (Section S6).

Active entries ($\mathrm{p}K_d > 6$, i.e. $K_d < 1$ μM) make up 16.7% of the Davis matrix. This 1 μM cutoff is a common activity threshold in kinase profiling. With the S-score's threshold sweep shifted up by one log unit (from $[5.5, 8.0]$ to $[6.5, 9.0]$), the S-score still correlates more with the entropy and Gini (0.88) than with the ratio (0.54).

The Klaeger study[10] profiled 243 clinical kinase inhibitors against the human kinome using Kinobeads chemical proteomics. Its Kinobeads table lists fitted targets for 229 of them. Apparent dissociation constants (K_d^{app}) were derived from dose–response curves across eight inhibitor concentrations in cancer cell lysates. Only high-confidence measurements (as annotated by the original authors) were retained, yielding 5,123 drug–target pairs for 222 drugs and 343 targets. Affinity values were converted to $\mathrm{p}K_d$ using the same formula as above. Missing entries, combinations for which no binding was detected, were assigned the background value $\mathrm{p}K_d = 5.0$. Measured values below the floor (194 entries, K_d^{app} above 10 μM) were set to the floor. The 343 targets are the proteins the Kinobeads pulldown quantified, including kinase-complex subunits, such as cyclins, captured with their kinases. We use the 217 targets that Klaeger et al. annotate as protein kinases and that are single genes. This leaves out lipid kinases, four protein groups (such as PRKCE;PRKCH), and Q6ZSR9, an uncharacterized protein without a kinase domain. It also leaves out BCR and TRRAP, which the source does not annotate as kinases and flags as possibly co-enriched

with ABL1 and CDK8. A table in the code repository lists every target of each dataset with the reason it was kept or removed. Results on the released panel are in Supporting Information (Section S6). Active entries ($pK_d > 6$) make up 4.5% of the Klaeger matrix. The 46 compounds profiled in both datasets (Discussion) are active against 15.0% of the Davis panel on average and 3.9% of the Klaeger panel, so the difference does not come from the compounds. These data cannot tell whether it comes from the chemoproteomic assay's detection threshold or from the two panels profiling different kinases.

The high-confidence filter removes 793 of the 5,916 drug–target rows (13.4%), and a removed pair is treated as one with no detected binding. Retaining the removed pairs shifts the Klaeger cells of Table 1 by at most 0.08 and leaves the four definitions' Klaeger panel-size p^* values unchanged (Supporting Information, Section S6).

To test whether our findings depend on the assay technology, we added two further datasets that measure kinase inhibition by different means. The Anastassiadis dataset[18] profiles 178 inhibitors against 300 kinases in a single-concentration (0.5 μM) radiometric activity assay, reported as percent residual kinase activity relative to a solvent control. We converted these to percent inhibition (100 minus residual activity, clipped to $[0, 100]$, where higher values denote stronger inhibition), with the 1.1% missing entries set to zero inhibition. Nine kinases are profiled in two complexes or isoforms (for example, CDK1 with cyclin A and with cyclin B). We keep the first-listed of each, leaving 291 targets. The Metz dataset[19] reports pK_i affinities for a large medicinal-chemistry campaign using the same negative-log scale as pK_d. The value $pK_i = 4.0$ is the lowest value reported anywhere in the released data (only 0.17% of measurements), so we assign it to unmeasured combinations as the floor. Coverage per compound is highly skewed (a median compound is measured against only 21 of the 172 kinases). Because of this, we restricted the dataset to the 704 compounds profiled against at least 50 of the 172 kinases ($\geq 29\%$ coverage). The threshold excludes the most sparsely profiled compounds while retaining a large enough compound set for the panel-size and instability analyses. Across the 704 retained compounds, coverage ranges from 29.1%

to 83.1% of the 172 kinases (mean 40.9%, median 37.2%, i.e. 64 of 172 kinases). Thus, for the median retained compound, most entries are floor-imputed, and fewer than a quarter of compounds have more than half of their entries measured (75th percentile 47.1%). A sizable minority sit close to the threshold: 23 of 704 (3.3%) have exactly 50 measured kinases, and 170 (24.1%) have 55 or fewer.

Floor imputation has a large effect on Metz. Nearly every measured value (99.9%) lies above the floor, so a compound measured against more kinases has more kinases with positive weight and looks less selective. Across the 704 compounds, the Spearman correlation between a compound's coverage and its median rank (rank 1 = most selective) is 0.98 for the entropy, 0.95 for the Gini, 0.76 for the S-score, and 0.11 for the ratio. Part of this is real, since compounds measured against more kinases are also active against a larger share of the kinases actually measured, but that association is much weaker ($r = 0.33$). No fully measured subset of useful size exists (only 7 kinases are measured for at least 80% of the compounds), so we keep Metz as a fourth dataset but read its distribution-based results, and their low agreement with the ratio, as largely reflecting coverage.

The four datasets span three assay technologies (competition binding, chemoproteomics, and functional activity) and two readout types (affinity and single-concentration inhibition). Each selectivity definition is computed on the native readout of its dataset, and every comparison between datasets uses rank correlations computed within each dataset. For the percent-inhibition and pK_i datasets, the S-score threshold and entropy/Gini baseline grids were shifted to span each dataset's active range (thresholds over $[50, 90]$% inhibition on Anastassiadis and p$K_i \in [5.0, 7.5]$ on Metz, and baselines over $[0, 30]$% and $[4.0, 5.5]$). The ratio's off-target index $k \in \{1, \ldots, 5\}$ needs no rescaling.

We excluded two further datasets. The Gao screen[20] has the same readout type as Anastassiadis, so it would add no new assay technology, and its raw data are not openly redistributable. The Patricelli study[21] profiles too few inhibitors for a ranking analysis.

Selectivity Definitions and Parameterization

We implemented four selectivity definitions, each with parameters whose values are not standardized in the literature. Throughout, a compound's binding profile is the vector of its affinities $\mathrm{p}K_{d,i}$ across the N panel kinases, with higher $\mathrm{p}K_d$ denoting tighter binding.

S-score. Karaman et al.[4] defined the selectivity score as the fraction of the panel that a compound binds: the number of kinases bound with a dissociation constant below a cutoff, divided by the number of distinct kinases tested. They reported $S(3\ \mu\mathrm{M})$ and $S(100\ \mathrm{nM})$, for K_d cutoffs of 3 μM and 100 nM. We implemented the same fraction, with the cutoff expressed as a threshold τ on $\mathrm{p}K_d$:

$$S(\tau) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[\mathrm{p}K_{d,i} > \tau]. \tag{1}$$

Here N is the number of kinases in the panel. On Metz this includes kinases a compound was not tested against, which are imputed at the floor and so count only in the denominator. We used $\tau \in \{5.5, 5.75, \ldots, 8.0\}$ (11 values), spanning K_d cutoffs from approximately 3 μM to 10 nM, which includes both of Karaman et al.'s cutoffs ($\tau \approx 5.5$ and $\tau = 7$).

Selectivity entropy. Uitdehaag and Zaman[5] treat the compound's binding across the panel as a probability distribution and report its Shannon entropy. A profile concentrated on few kinases has low entropy and is selective. In the original definition, the weight on each kinase is its association constant $K_{a,i} = 1/K_{d,i}$ (the share of compound that would be bound to kinase i in an equimolar mixture), so $p_i = K_{a,i} / \sum_j K_{a,j}$. We implemented the entropy on the same log-affinity scale used by the other metrics, weighting each kinase by its affinity in excess of a baseline β:

$$H(\beta) = -\sum_{i=1}^{N} p_i \log_2 p_i, \quad p_i = \frac{\max(\mathrm{p}K_{d,i} - \beta,\, 0)}{\sum_j \max(\mathrm{p}K_{d,j} - \beta,\, 0)}, \tag{2}$$

where β is the affinity below which binding is treated as negligible. This makes β an explicit parameter. We swept $\beta \in \{5.0, 5.25, \ldots, 6.5\}$ (7 values), a narrower range than the S-score's τ sweep. Raising either parameter leaves a growing share of compounds with no kinase above it (on Klaeger, 9.9% of compounds at $\mathrm{p}K_d = 6.0$ and 58% at $\mathrm{p}K_d = 8.0$), where the S-score becomes 0 and the entropy and Gini become undefined. The τ range is wider because it was chosen to include Karaman et al.'s cutoffs. The baseline β has no published reference points and was kept near the $\mathrm{p}K_d = 6$ activity threshold. We compare this weighting with the original association-constant weighting below.

Gini coefficient. Graczyk[6] adapted the Gini coefficient from economics: kinases are sorted by activity and the cumulative fraction of total binding is plotted against the cumulative fraction of kinases (a Lorenz curve). The Gini coefficient, which measures how unequally binding is distributed across the kinome, is twice the area between this curve and the diagonal of equal distribution. The original work computes it from percent-inhibition values. As with the entropy, we computed it on the baseline-subtracted log-affinity profile, using the standard discrete formula:

$$G(\beta) = \frac{2\sum_{i=1}^{N} i \cdot x_{(i)}}{N\sum_{i=1}^{N} x_{(i)}} - \frac{N+1}{N}, \qquad x_{(i)} = \max(\mathrm{p}K_{d,i} - \beta,\, 0) \text{ sorted ascending.} \tag{3}$$

We swept β over the same range as for the entropy. A compound whose binding is concentrated on one kinase has a highly unequal distribution (G is near 1), while a compound that binds many kinases equally has G near 0. Higher values therefore indicate greater selectivity.

Ratio. Classical pharmacological selectivity is pairwise. It quantifies how much more tightly a compound binds its primary target than an off-target. Karaman et al.[4] also reported a score of this kind alongside the S-score: the fraction of tested kinases bound within tenfold of the affinity for the compound's intended primary target, which combines a ratio cutoff with an S-score-style count. Anchoring on an annotated primary target avoids the ratio's near-tie

instability (Type 2, Results). The score, however, has its own unstandardized parameter (the tenfold cutoff) and, as a fraction of the panel, violates D3 in the same way as the S-score. We instead implemented the pairwise comparison from the binding profile alone, as every measure compared here is a function of the profile (Motivated Properties, below). We included the reference off-target as an explicit parameter whose effect can be studied directly: the fold-selectivity, in log units, between the primary target and the k-th ranked off-target:

$$R(k) = \mathrm{p}K_{d,(1)} - \mathrm{p}K_{d,(k+1)} = \log_{10} \frac{K_{d,(k+1)}}{K_{d,(1)}}, \tag{4}$$

where $\mathrm{p}K_{d,(j)}$ is the j-th largest affinity in the profile and $k \in \{1, 2, 3, 4, 5\}$ selects the reference off-target. We capped k at 5 because, on the sparser Klaeger panel, a large share of compounds already have fewer than five active kinases (median 4; 51% have fewer than five). For those compounds, a larger k would compare the primary target against the assay floor instead of an off-target. Bosc et al.'s ranking score[7] is the same difference taken at a chosen rank, so $R(k)$ is their ranking score at rank $k + 1$. On Anastassiadis, $R(k)$ is a difference in percent inhibition rather than a log fold-selectivity.

Our $R(k)$ scores every compound, substituting the value assigned to non-detection for a missing off-target. Bosc et al.'s ranking score also scores every compound, on a matrix with non-binding pairs set to 10.1 μM, so $R(k)$ shares its D1 violation with the published score.

The concentration- and target-dependent selectivity (CATDS) score of Klaeger et al.[10] is not a ratio in this sense, since it normalizes over the whole panel (Background). A proxy computed from our profiles correlates more with the Gini than with $R(k)$ on every dataset (Discussion, Supporting Information Table S2), but it does not reproduce the concentration dependence of the published score. We therefore represent the ratio family by $R(k)$ alone.

Comparators. Three further measures are used in the property and recovery analyses. The active count is the number of targets above the activity threshold τ^* (below), negated so that higher values are more selective. The second is the entropy with its baseline at the activity

threshold, $\beta = \tau^*$, which gives no weight to binding below τ^*. The third is the S-score with its threshold at the floor x_0 (below), the share of the panel with detected binding. All three are scored only for compounds with a target above τ^*, as D1 (below) requires.

Faithfulness to the original definitions. The S-score and the ratio are computed in the form given by their original sources (a panel-hit fraction and a primary-to-off-target affinity ratio, respectively). For the entropy and the Gini coefficient, we used the baseline-subtracted log-affinity profile in place of the original association constants (entropy) and percent-inhibition values (Gini), so that all four metrics operate on one common scale. The baseline β is therefore a parameter of our operationalization, not of the published definitions. Unless stated otherwise, every entropy and Gini result in this paper concerns this form.

To test how much this choice matters, we also computed the entropy in its published form, with association-constant weighting ($p_i = K_{a,i} / \sum_j K_{a,j}$), and the Gini of the association constants as an affinity-scale analogue. Two choices are involved. The first is the weighting. The second is the treatment of pairs coded at the floor: pairs with no detected binding on Davis and Klaeger, and pairs not tested on Metz. The log-affinity forms with $\beta = x_0$ give these pairs no weight. Association constants give them a small positive weight if they are kept at the floor value. We report both treatments, floor-coded pairs kept at the floor and given no weight. Neither is free of assumptions, since a floor-coded pair has no measured affinity. On Anastassiadis, the one dataset with a native percent-inhibition scale, we also computed the Gini without baseline subtraction. Results compares these forms with the log-affinity forms (Figure 1; Supporting Information, Table S3).

Rank Stability Analysis

For each definition and each parameter value, compounds were ranked from most selective (rank 1) to least selective (rank n, where n is the number of compounds in the dataset). The S-score and the raw Shannon entropy both run opposite to this convention (a higher fraction

of panel hits, or a higher raw entropy, indicates a *more* promiscuous compound), so both are negated before ranking throughout this paper. The Gini coefficient and $R(k)$ already increase with selectivity and are used unnegated. Ties were broken by average rank. Where the entropy and Gini are undefined (no kinase above the baseline β), our implementation sets both scores to 0. This ranks the compound as maximally selective under the (negated) entropy measure, tied with single-target profiles, and as minimally selective under the Gini. A compound can be undefined at some baselines of the sweep and defined at others. Results compares this with computing instability only over the configurations in which a compound's score is defined. The ratio substitutes the value assigned to non-detection for a missing off-target (above): $\mathrm{p}K_d = 5.0$ on Davis and Klaeger, $\mathrm{p}K_i = 4.0$ on Metz, and 0% inhibition on Anastassiadis. The S-score is defined for every compound. This yielded 30 rank vectors per dataset for Davis, Klaeger, and Metz (11 for the S-score, 7 for entropy, 7 for Gini, and 5 for ratio), and 28 for Anastassiadis, whose coarser native percent-inhibition scale supports only 9 S-score threshold values rather than 11.

For each compound, rank instability was quantified as the standard deviation of its rank across all configurations for that dataset (overall instability) and separately across parameter configurations within each definition family (within-family instability). Cross-definition-family agreement was assessed using Spearman rank correlation between the median rank vectors of each definition family. The median across parameter values was used because it is less sensitive to outlying configurations than the mean. Using mean ranks instead changes no correlation by more than 0.08 on any dataset.

Associations between rank instability and binding profile features were assessed using Spearman correlation on Klaeger and Davis. Both report $\mathrm{p}K_d$, so the features are on the same scale and each relation found on one can be tested on the other. Profile features computed for each compound included: the number of active kinases ($\mathrm{p}K_d > 6$), the gap between the top and second-ranked affinities, the standard deviation and range of active affinities, the absolute affinity of the second-ranked target, and the skewness of the active affinities. The first two

were also tested on Anastassiadis and Metz, against entropy and ratio instability respectively (Results). These associations are exploratory. Five further features were examined (Results), and correction for multiple testing covers only the 24 tests of Table 2. All correlations are reported with two-tailed p-values, uncorrected for multiple testing unless stated. Significance thresholds of $p < 0.05$, $p < 0.01$, and $p < 0.001$ are denoted *, **, and *** respectively. Analyses were performed in Python 3 using NumPy, pandas, and SciPy.[22–24] A secondary analysis relating selectivity rankings to clinical adverse-event reports is summarized in the Limitations section. The analysis code and full per-definition results are available in the code repository (see Data and Software Availability), and Supporting Information Table S7 lists the script that computes each reported result.

Motivated Properties

The instability patterns reported in Results motivate three properties. Each is stated formally below, together with the failure mode it targets and how a measure is assessed against it. We evaluate the four definitions, the two comparators and the candidate measure against them (Results). Let $\mathbf{x} = (x_1, \ldots, x_N) \in \mathbb{R}^N$ denote a compound's binding profile across N kinases, where $x_i = \mathrm{p}K_{d,i}$ and N is the panel size, which varies across datasets and subpanels. A selectivity measure is a function $f : \mathbb{R}^N \to \mathbb{R}$, defined for every panel size N, where higher values indicate greater selectivity. A ranking over a set of compounds is defined by ordering their f-scores. We write x_0 for the assay floor ($\mathrm{p}K_d = 5.0$ on Davis and Klaeger, $\mathrm{p}K_i = 4.0$ on Metz, and 20% inhibition on Anastassiadis, where 54% of entries lie between 0% and 20%, so a floor at 0% would admit assay noise) and τ^* for the activity threshold ($\mathrm{p}K_d$ or $\mathrm{p}K_i = 6$; 50% inhibition on Anastassiadis).

D1 (reliability threshold). $f(\mathbf{x})$ is undefined, or explicitly flagged as unreliable, whenever $\max_i x_i \leq \tau^*$.

A compound with no binding above the activity threshold has no target against which to judge selectivity. Its rank then depends on how each measure handles such profiles (Type 1,

Results). D1 is a reporting rule and is assessed as such.

D2 (distributional consistency). Let $B = [x_0, \tau^*]$, the interval between the floor and the activity threshold, where an analyst must decide how much sub-threshold binding counts. Let θ be a parameter of f that an analyst must choose on the affinity scale (the S-score's τ and the entropy's and Gini's β, which place the activity boundary, or the level below which the ratio replaces an off-target affinity). A pair of compounds $\mathbf{x}, \mathbf{y}$ *reverses in* B if $f(\mathbf{x}; \theta) - f(\mathbf{y}; \theta)$ is positive at some $\theta \in B$ and negative at another. D2 requires that at most a fraction ε of the pairs of compounds that D1 allows to be scored reverse in B.

We use $\varepsilon = 1\%$. D2 addresses the baseline sensitivity of compounds with few active kinases (Results) and is assessed empirically on each dataset. The S-score is evaluated at every observed affinity in B, where its value changes, and the other parameters on 201 equally spaced values (101 values change no fraction by more than 0.1 percentage point). Fixing a parameter before analysis is a protocol decision. It does not make a ranking robust to the parameter, and D2 reports how much the ranking would change had a different value been chosen.

D3 (monotonicity under weak off-target addition). For any $\mathbf{x}$ with $\max_i x_i > \tau^*$, if $\mathbf{x}' = (x_1, \ldots, x_N, x_{N+1})$ with $x_0 < x_{N+1} \leq \tau^*$, then $f(\mathbf{x}') \leq f(\mathbf{x})$.

That is, adding a kinase that the compound binds above the floor but below the activity threshold must not increase its score. Otherwise, a compound can be made to look more selective by profiling it against more kinases it barely binds. D3 is assessed from each measure's functional form.

We state no property for the ratio's choice of reference off-target. Changing which kinases count as tolerated co-targets changes the question a selectivity score answers, so scores for different references need not agree. The ratio's dependence on k is reported as an empirical result (Type 2, Results).

Recoverability. Separately from the properties, we evaluate how well each measure's ranking is recovered from smaller panels. Subpanels of m targets are drawn uniformly at

random from the curated targets. Two compounds are clearly separated if their full-panel ranks are at least a quarter of the scored compounds apart. Recoverability at size m is the share of clearly separated pairs that are scored on the subpanel and keep their order, with a pair tied on the subpanel counted as one half. It therefore combines ordering with coverage, the share of compounds still scored, and we also report each separately. Recoverability is a criterion for this sampling design, not a property that every profile can meet: a subpanel that omits all of a compound's active targets cannot recover its rank.

A Candidate Measure

The candidate measure is an entropy-based score that meets D1 and D3 and weights binding between the floor and the activity threshold continuously in affinity. The active count and the entropy with $\beta = \tau^*$ meet D3 by giving such binding no weight, and the S-score with $\tau = x_0$ by counting every kinase above the floor equally. For a profile $\mathbf{x}$ with $\max_i x_i > \tau^*$ (undefined otherwise, satisfying D1), define

$$w_i = \mathbf{1}[x_i > x_0]\, T \log\bigl(1 + e^{(x_i-\beta)/T}\bigr), \qquad p_i = \frac{w_i}{\sum_j w_j}, \qquad f(\mathbf{x}) = \sum_i p_i \log_2 p_i, \tag{5}$$

where x_0 is the assay floor, β is a smooth emphasis baseline (set to x_0 in the results below), and T is a smoothing width in the native units of the profile. We set $T = \tau^* - x_0$, the distance from the floor to the activity threshold: 1 pK_d unit on Davis and Klaeger, 2 pK_i units on Metz, and 30 percentage points on Anastassiadis, whose native scale is 0–100% inhibition rather than log-affinity. The function f is the negative Shannon entropy of the smoothed, normalized profile, i.e. the negative logarithm of the effective number of targets the compound binds.

The candidate differs from the selectivity entropy in two ways. First, it is undefined for a compound with no active target, as D1 requires. Second, it replaces the entropy's hard cutoff $\max(x_i - \beta, 0)$ with a softplus, for D3. The softplus is applied only to kinases above

the floor x_0, since it would otherwise give weight to non-binders. The hard cutoff gives a kinase just above β a weight close to zero, and this is what lets a sub-threshold addition raise the entropy's score. The softplus gives every gated kinase a weight of at least $T \ln 2$, which rules this out:

Proposition. With $\beta = x_0$ and $T \geq (\tau^* - x_0)/\ln(2^e - 1) \approx 0.58\,(\tau^* - x_0)$, adding a kinase with affinity in $(x_0, \tau^*]$ never raises the candidate's score.

Proof. Every gated kinase, the added one included, has weight above $m = T \ln 2$. Since the softplus increases with affinity, an added kinase with affinity at most τ^* has weight $a \leq T \ln(1 + e^{(\tau^* - x_0)/T})$, and, by the assumption on T in the proposition, this is at most $e\,m$.

Let the original profile have $n \geq 1$ gated kinases ($n \geq 1$ because D1 scores it only if some kinase lies above $\tau^* > x_0$), total weight $W > nm$, and entropy H (in nats), and let $q = a/(W + a)$ be the share of the total weight carried by the added kinase. Once that kinase is added, the entropy is $(1 - q)H + h(q)$, where $h(q) = -q \ln q - (1 - q) \ln(1 - q)$ is the binary entropy. So, the entropy does not decrease if and only if $H \leq h(q)/q$. Because h is concave with $h(0) = 0$, $h(q)/q$ decreases in q. Since $q < r/(n + r)$ with $r = a/m \leq e$, this gives $h(q)/q > \ln \frac{n+r}{r} + \frac{n}{r} \ln \frac{n+r}{n}$. As $H \leq \ln n$, it suffices that $\ln n$ is at most this bound, which rearranges to $\ln r \leq (1 + \frac{n}{r}) \ln(1 + \frac{r}{n})$. The right-hand side is at least 1 for all $n, r > 0$, because $\ln(1 + t) \geq t/(1 + t)$, and $\ln r \leq 1$. The entropy therefore does not decrease, so the score (its negative) does not increase. □

The choice $T = \tau^* - x_0$ satisfies this condition on every dataset. With this T, the same argument covers every $\beta \in B$: the lightest gated kinase then has weight above $m = T \ln(1 + e^{(x_0 - \beta)/T})$, and $a/m \leq 2.22 < e$.

The gate makes the candidate discontinuous at x_0: a kinase just above the floor carries weight $T \ln 2$, and one at the floor carries none. The D3 guarantee depends on this minimum weight, not on the smoothness of the hinge. Measurement noise of 0.3 pK units (9 percentage points on Anastassiadis) changes which compounds D1 scores for at most 11 compounds in any draw. Over the compounds scored with and without noise, the candidate's ranking

stays at Spearman 0.978 or higher with the noise-free ranking, against 0.983 or higher for the entropy with $\beta = x_0$.

Results

Agreement Between Definitions Depends on Analysis Choices

Three analysis choices change how the definitions' median rankings relate (Table 1): which compounds are included, how the entropy weights affinities, and how pairs coded at the floor are treated.

With the entropy and Gini in their log-affinity forms, the S-score, entropy and Gini agree more with one another than with the ratio among compounds with an active kinase. On Davis, where every compound has one, they correlate at 0.963–0.999, and the ratio with each at 0.51–0.56. Among the 200 Klaeger compounds with an active kinase, they correlate at 0.89–0.997, and the ratio with each at 0.38–0.64. With all 222 Klaeger compounds included, the Gini correlates more with the ratio (0.66) than with the S-score (0.51). The difference comes from the 22 compounds with no active kinase, which each definition scores by its own convention (Type 1, below). We tested whether the Gini sits closer to the S-score than to the ratio by bootstrapping the difference between its two correlations (1,000 resamples of compounds with replacement). With all compounds included, the difference is not significant (95% interval $[-0.35, 0.06]$). Among the 200 compounds with an active kinase, the Gini sits significantly closer to the S-score (0.89 against 0.64, interval $[0.16, 0.36]$). On the log-affinity scale, the S-score, entropy, and Gini all quantify the concentration of activity across the full panel, while the ratio quantifies the gap between the primary target and one off-target and ignores the rest of the profile.

In its published form, with association-constant weighting, the entropy correlates most with the ratio on Davis and Klaeger, whichever compounds are included and whether floor-coded pairs are kept at the floor or given no weight (Table 1). The same holds on Metz

Table 1: Spearman correlations between median rankings by definition. S: S-score; E: entropy; G: Gini; R: ratio, with the entropy and Gini in their log-affinity forms. E_{K_a}: the published entropy, with association-constant weighting, with floor-coded pairs given no weight and, in parentheses, kept at the floor. Every Davis compound has an active kinase.

	Davis	Klaeger	
Pair	all ($n = 68$)	all ($n = 222$)	active ($n = 200$)
S–E	0.963	0.896	0.897
S–G	0.963	0.508	0.892
E–G	0.999	0.658	0.997
S–R	0.563	0.078	0.378
E–R	0.513	0.344	0.622
G–R	0.510	0.655	0.635
E_{K_a}–S	0.738 (0.344)	0.571 (−0.597)	0.564 (−0.412)
E_{K_a}–G	0.736 (0.368)	0.530 (0.077)	0.782 (−0.094)
E_{K_a}–R	0.861 (0.688)	0.657 (0.648)	0.916 (0.591)

(Supporting Information, Table S3). How far it is from the S-score and Gini depends on the floor-coded pairs, which make up 71% of the Davis matrix. On Davis, its correlation with the S-score is 0.34 with these pairs at the floor and 0.74 with them given no weight. Against our log-affinity forms, the published entropy correlates at $r = 0.37$ (Davis) and -0.31 (Klaeger) with floor-coded pairs at the floor, and at 0.74 and 0.79 with them given no weight (Figure 1). For the Gini of the association constants, the corresponding values are 0.01 and -0.20, and 0.83 and 0.71. On Anastassiadis, the native percent-inhibition Gini correlates with the entropy ($r = 0.63$), and weakly and about equally with the S-score (0.34) and the ratio (0.33).

The two-family structure is therefore a property of the log-affinity operationalization and of the compounds included. It does not hold for the entropy as originally published. The weighting and the treatment of floor-coded pairs are themselves sources of definitional instability, in addition to the three characterized below.

Agreement on Anastassiadis and Metz

Among compounds with an active kinase, the pattern extends to the two further datasets, across three assay technologies and both readout types (Figure 2). The S-score, entropy, and Gini remain mutually correlated on all four datasets (Spearman $r = 0.81$–0.999), while the ratio is the outlier. With all compounds included, its correlation with the distribution-based family is 0.48–0.55 on the Anastassiadis functional-activity dataset and only 0.14–0.19 on the Metz pK_i dataset. On Anastassiadis, the separation comes mainly from the 45 compounds with no active kinase (Type 1, below). Among the other 133, the ratio's correlation with the family rises to 0.79–0.84, against 0.87–0.999 within it. Metz shows the effect of missing measurements. A median compound is measured against 64 of the 172 kinases, the rest are set to the floor, and the entropy and Gini rankings largely follow how many kinases were measured (Methods). Metz therefore shows how floor imputation changes rankings, not how the definitions relate on complete profiles.

The instability findings hold on these datasets, with qualifications. Pooled across the definitions' sweeps, compounds with no active kinase have higher rank instability than active compounds wherever they occur (mean rank $\sigma = 36.6$ vs. 21.8 on Anastassiadis and 178.5 vs. 108.1 on Metz; for Klaeger see D1, below). On each dataset the contrast comes mainly from the definitions disagreeing with one another rather than from each definition's own parameter sweep (Type 1, below). Within a single sweep, it is significant for the Gini, the entropy and the ratio on Anastassiadis. It reverses for the S-score, the entropy and the Gini on Metz ($\sigma = 7.8$ vs. 71.2, 11.0 vs. 24.5 and 5.4 vs. 24.2; all $p < 0.05$). The number of active kinases correlates negatively with entropy instability on every dataset ($r = -0.30$ to -0.42; on Davis only nominally, $p = 0.01$). The top_1–top_2 affinity gap is associated with ratio instability on Davis, Klaeger and Metz ($r = -0.44$, -0.34, and -0.27, all $p < 0.001$), and on Anastassiadis once the compounds saturated at the 100% inhibition ceiling are excluded (below).

Rank Instability Is Structured and Associated with Profile Shape

Overall rank instability varies substantially across compounds (Klaeger: mean $\sigma = 36.7$, range 8.7–94.0; Davis: mean $\sigma = 7.5$, range 2.1–16.4). Instability is not uniformly distributed: within each definition family, specific binding profile features are associated with which compounds receive unstable rankings (Table 2). These associations are exploratory (Methods).

Within the ratio family, the gap between the top and second-ranked affinities (top_1–top_2 gap) is the feature most strongly associated with instability on both datasets (Figure 3; Type 2, below).

Within the entropy family, the number of active kinases ($\text{p}K_d > 6$) correlates negatively with instability (Table 2; for the Gini, see below). Compounds with many active kinases have broad profiles whose entropy scores are stable across baseline choices. Compounds with few active kinases are sensitive to the baseline because small changes can move targets in or out of the active set, altering the distribution substantially. Consistent with this, the share of a compound's above-floor kinases that cross some baseline in the sweep correlates with its entropy instability ($r = 0.21$, $p = 0.004$ on Klaeger). The active affinity range is also associated with S-score and entropy instability on Klaeger ($r = -0.37$ and -0.25, both $p < 0.001$). For the Gini it is not ($r = -0.05$, $p = 0.45$). Range rises with the number of active kinases (Spearman 0.76). Holding that number fixed, range is still associated with S-score instability (partial $r = -0.28$, $p < 0.001$) and no longer with entropy instability ($r = 0.08$, $p = 0.24$).

Gini instability's strongest correlate on Klaeger ($r = +0.262$, $p < 0.001$) is skewness. Unlike the other features above, it describes the shape of the active-affinity distribution, whose inequality the Gini measures. Right-skewed active distributions are the least stable. In such profiles most active kinases sit just above the threshold, with one or a few much more potent outliers. A shift in β therefore moves much of the weight in or out of the active set, which accounts for part of the association (Supporting Information, Section S6). The relationship does not replicate on Davis ($r = +0.176$, $p = 0.15$).

Unlike the other families, the S-score's instability correlates significantly with every profile-shape feature except the top_1–top_2 gap (Table 2), most strongly with the range of active affinities. This S-score pattern is specific to Klaeger: on Davis, no profile feature is significantly associated with S-score instability (all $|r| \leq 0.21$, Table 2, right). On the 46 compounds profiled in both, the association with active affinity range is weak on both datasets (Klaeger $r = -0.28$, $p = 0.09$, for the 37 with at least two active kinases; Davis $r = -0.16$, $p = 0.30$), so these data cannot tell whether the difference comes from the compounds or the assays (the same compounds have a median of 4 active kinases on Klaeger and 32 on Davis). Two relations in Table 2 are significant on both pK_d datasets: the ratio's dependence on the top_1–top_2 gap and the entropy's dependence on the number of active kinases. With a Bonferroni correction for the 24 tests per dataset, only the first remains significant on both. On Anastassiadis the ratio–gap dependence fails ($r = -0.13$, $p = 0.13$) because of eight compounds whose two best-inhibited kinases are both at 99.5% inhibition or more: the single-concentration readout saturates at 100%, so their top_1–top_2 gap is unresolved. Excluding them, the dependence holds there too ($r = -0.31$, $p < 0.001$). For the Gini, the dependence on the number of active kinases is significant on Davis ($r = -0.31$) and not on Klaeger ($r = -0.07$).

Table 2: Spearman correlations between binding profile features and within-family rank instability, on the Klaeger ($n = 200$ active drugs) and Davis ($n = 68$) datasets. Columns are the S-score (S), entropy (E), Gini (G), and ratio (R) families. Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

	Klaeger ($n = 200$)				Davis ($n = 68$)			
Feature	S	E	G	R	S	E	G	R
n_{active}	-0.253^{***}	-0.394^{***}	-0.075	-0.025	-0.210	-0.301^{*}	-0.313^{**}	$+0.007$
top_1–top_2 gap	$+0.016$	$+0.057$	$+0.057$	-0.342^{***}	$+0.043$	$+0.074$	$+0.058$	-0.445^{***}
Active affinity std	-0.308^{***}	-0.132	-0.056	-0.121	-0.017	$+0.020$	$+0.035$	-0.176
Active affinity range	-0.370^{***}	-0.247^{***}	-0.054	-0.113	-0.023	-0.065	-0.076	-0.240^{*}
top_2 pK_d	-0.300^{***}	-0.230^{**}	-0.219^{**}	$+0.221^{**}$	-0.174	-0.186	-0.190	$+0.147$
Active affinity skew	-0.236^{***}	-0.096	$+0.262^{***}$	-0.107	$+0.041$	$+0.195$	$+0.176$	-0.113

The skewness of a compound's active affinities is undefined when it has fewer than three active kinases, and so are their standard deviation and range when it has fewer than two.

Table 2 assigns these compounds the value 0 (the vertical clusters at zero in Figure 3). Recomputing each correlation over only the compounds for which the feature is defined strengthens every Klaeger association reported above (Gini–skewness $r = +0.381$; range against S-score and entropy instability $r = -0.466$ and -0.417, all $p < 0.001$) and changes no Davis correlation materially.

Five further features, including two concentration indices from industrial-organization economics, were either near-collinear with n_{active} or weaker than the strongest feature in Table 2 (Supporting Information, Section S6).

Three Structurally Distinct Sources of Instability

Examination of the most unstable compounds in the Klaeger dataset reveals three structurally distinct sources of rank instability, each implicating a different aspect of how the definitions are constructed. All 20 most unstable Klaeger compounds have no active kinase (Type 1). Among the 20 most unstable compounds with an active kinase, five have a top_1–top_2 gap below 0.1 (Type 2) and eight have at least 10 active kinases (Type 3). Figure 4 shows example profiles from Davis (the extremes are easier to see on the complete matrix there).

Type 1: Zero-active compounds. The most unstable Klaeger compounds all have zero active kinases at the $\text{p}K_d > 6$ threshold. Six of them (AZD-2014, AZD-8055, AZD-8186, BMS-911543, SGX-523, and volitinib; rank $\sigma = 94.0$) show no binding to any protein kinase on the panel. Their zero activity is an assay result: all six are established inhibitors of specific targets (mTOR, mTOR, PI3Kβ, JAK2, MET, and MET, respectively), none of which appear as high-confidence hits in the Klaeger chemoproteomic screen. For three of them (AZD-2014, AZD-8055, AZD-8186), the target was not quantified for any of the 229 compounds with a fitted target in the Kinobeads table. This is consistent with low expression or poor capture of those kinases in the cell lysates. For the other three, the target was quantified for other compounds: JAK2 for 7 and MET for 32 (up to $\text{p}K_d = 8.3$ and 9.5). Absence of the target from the lysate therefore does not explain why no binding was detected

for BMS-911543, SGX-523, or volitinib. These data cannot tell whether compound-specific capture, competition or curve fitting does.

All four definitions nonetheless score the 22 zero-active compounds from sub-threshold values alone, and they disagree about them. The S-score and entropy rank them among the most selective (median ranks 24 and 20 of 222), and the ratio near the bottom (182). The Gini ranks them near the top where its score is defined, and near the bottom where the convention for undefined scores applies (Methods; mean ranks 32 and 212). Their pooled rank instability is therefore high (D1, below). Davis has no such compounds.

Type 2: Near-tied top targets. When an active compound's two best-bound kinases have near-identical affinities, the ratio depends heavily on k. $R(1)$ compares the two near-tied kinases and is close to zero, and $R(2)$, which skips past the tie, can be large. On the 22 Klaeger compounds with an active kinase and a top_1–top_2 gap below 0.05, mean $R(1) = 0.02$ against mean $R(2) = 0.50$. The two values answer different questions: $R(1)$ treats the second kinase as an off-target, and $R(2)$ treats it as a co-target. On Davis, fedratinib, with JAK2 and GAK tied, has the most unstable ratio ranking (Figure 4, right).

Type 3: Broad profiles, often with one dominant target. Among active compounds, the ten largest entropy-ratio disagreements in the Klaeger dataset all involve profiles with 9–27 active kinases. Seven of them also have one clearly dominant target (top_1–top_2 gap above 1), and the ratio ranks these as far more selective than the entropy and Gini do (Supporting Information, Section S6).

Which compounds have the most stable overall ranking differs between datasets: on Klaeger the most promiscuous, and on Davis, on average, those with few active kinases (Supporting Information, Section S6).

Existing Definitions Against the Properties

Table 3 summarizes how the four definitions, three comparators, and the candidate measure fare against the properties D1–D3 defined in Methods. None of them satisfies D2: moving

the activity boundary across B reorders compounds under every measure in the table. D1 and D3 separate the measures. Simple constructions satisfy both: the active count, the entropy with $\beta = \tau^*$, and the S-score with its threshold at the floor, each reported only for compounds with an active target.

Table 3: Satisfaction of the three motivated properties by the existing selectivity definitions, three comparators, and the candidate measure. ✓: satisfied. ×: violated. D1 is assessed from each measure's reporting rule. The comparators (S-score with $\tau = x_0$, count, entropy with $\beta = \tau^*$) are reported only for compounds with an active target, and any measure can adopt this gate. D2 is assessed empirically, with the parameter that places the activity boundary swept over B; a comparator is assessed with the parameter it fixes swept. A measure satisfies D2 only if at most ε of pairs reverse on every dataset (for the ratio, for every k). D3 is assessed from the functional form. With the candidate's gate fixed, its emphasis baseline stays within ε.

	S-score		Count	Entropy		Gini	Ratio	Candidate
	$\tau > x_0$	$\tau = x_0$		$\beta = x_0$	$\beta = \tau^*$			
D1 (reliability)	×	✓	✓	×	✓	×	×	✓
D2 (consistency)	×	×	×	×	×	×	×	×
D3 (monotonicity)	×	✓	✓	×	✓	×	✓	✓

D1. None of the four definitions has a reliability condition, so each scores the 22 Klaeger compounds with no binding above the activity threshold. Pooled across the four definitions' sweeps, their rank instability is $\sigma = 77.9$ against 32.2 for active compounds ($p < 0.001$, Mann–Whitney U test; Figure 5, left). Within a single definition the contrast is significant only for the Gini (65.0 against 16.3), through its convention for undefined scores (Methods). It is small and not significant for the entropy (11.9 against 12.6, $p = 0.91$) and the ratio (17.5 against 19.1, $p = 0.51$), and reverses for the S-score (19.0 against 25.7, $p = 0.01$). The pooled contrast persists with the undefined entropy and Gini configurations dropped (66.2 against 31.7) and with only the S-score and ratio pooled (70.6 against 37.7, both $p < 0.001$). Computed only over each compound's defined configurations, zero-active compounds are *more* stable than active ones within the entropy (7.0 against 13.5) and the Gini (6.7 against 13.6, both $p = 0.002$). The case for D1 is therefore not that these scores are unstable within a definition, but that they rest on sub-threshold values alone, which each definition treats by

its own convention. Any of the four definitions could satisfy D1 by withholding or flagging such scores, as Bosc et al. did for the S-score.

D2. Swept over $B = [x_0, \tau^*]$, the S-score's threshold reverses the order of 8.4–13.0% of pairs of scored compounds on Davis, Klaeger and Anastassiadis, and 21.8% on Metz. The entropy's and Gini's baselines reverse 7.9–12.4%, and 22.5–22.8% on Metz. Each comparator fixes a parameter of one of these sweeps and so shares its sensitivity: the S-score with $\tau = x_0$ and the active count reverse the same pairs as the S-score, and the entropy with $\beta = \tau^*$ the same pairs as the entropy. The candidate's gate, swept with T held at $\tau^* - x_0$, reverses 9.2–14.2%, and 23.6% on Metz. On Klaeger its per-compound instability is similar to that of the entropy's baseline (mean rank $\sigma = 7.4$ and 7.5; Figure 5, right). For the ratio, the parameter is the level b in $R_b(k) = x_{(1)} - \max(x_{(k+1)}, b)$, which replaces any off-target affinity below b, measured or not, by b. Moving b across B reverses 2.1–13.7% of pairs on Klaeger and Anastassiadis, depending on k, 0–2.6% on Davis and 0.04–1.03% on Metz (Supporting Information, Table S4). It exceeds ε for every k on Klaeger and Anastassiadis. None of the measures in Table 3 therefore satisfies D2, and the activity boundary must be fixed in the analysis protocol, with its effect reported. The published entropy has no such boundary parameter and is not assessed against D2. With its gate fixed, the candidate's remaining parameter, its emphasis baseline, reverses at most 0.72% of pairs.

D3. The S-score violates D3 for any threshold above the floor. Adding a kinase bound between the floor and the threshold increases N but not the number of hits, so the hit fraction falls and the compound looks more selective. With the threshold at the floor, every addition that D3 allows is a hit. A compound with C hits among N kinases then moves from $-C/N$ to $-(C+1)/(N+1)$, which is not higher, so the S-score with $\tau = x_0$ satisfies D3. The active count ranks every panel exactly as the S-score at τ^* does but does not divide by N, so it also satisfies D3. D3 therefore concerns how scores change when a panel is extended, not how compounds are ranked on a given panel. This matters whenever panels differ in size or composition, as between Davis and Klaeger. The Gini violates D3 by construction: it is

computed over the full panel, so adding a weakly bound kinase can raise its inequality. The ratio satisfies D3, because a sub-threshold kinase cannot displace the primary target and cannot lower the reference off-target's affinity. The entropy satisfies D3 at $\beta = \tau^*$, where no addition below τ^* carries weight, and for $\beta < \tau^*$ only for additions at or below β ("Why the Smooth Hinge Matters", below).

The Candidate Measure

The candidate measure (Methods) is correctly oriented: its score falls with the number of active kinases on every dataset (Spearman -0.78 to -0.97). The non-binder gate is what makes this so. Without it, the non-binding kinases at the floor dominate the profile and the score inverts, rising with the number of active kinases ($r = +0.91$ on Klaeger).

Because the candidate changes only the entropy's gate and hinge, the two rank the compounds they both score almost identically (Spearman 0.994–0.998 on all four datasets). The candidate satisfies D1 by its reliability gate and D3 by its functional form (Methods, Proposition), while still weighting binding between the floor and the activity threshold. It also ranks compounds almost identically to the S-score with $\tau = x_0$ (Spearman 0.998–0.9997), which satisfies both properties too. What the candidate adds is a weight on each target that increases continuously with affinity above the floor, which separates compounds that a count of detected targets ties. Like every other measure, it fails D2 when its gate is moved (above). The activity threshold τ^*, which sets its D1 gate and T, is also fixed. Moving it from 5.5 to 7 (40% to 70% on Anastassiadis) changes which compounds are scored but barely changes how the candidate ranks those scored at both values (Spearman ≥ 0.999 on every dataset). D3 also holds for every scored compound in all four datasets when one kinase is added at τ^*, the strongest binding that D3 allows (Supporting Information Table S4).

Variants of the candidate behave in much the same way. Sweeping the emphasis baseline reverses at most ε of pairs for Rényi orders 0.5 and 1 on Davis and Klaeger, and for smoothing widths from 0.75 to 2 times $\tau^* - x_0$ on Klaeger. Order 2 stays within ε on Davis and slightly

exceeds it on Klaeger (1.04%). D3 is proved only for the Shannon order, $\alpha = 1$. These variants are reported in Supporting Information (Section S5).

Recovery from Small Panels

Karaman et al.[4] scored each compound's S-score on random subpanels and found that it varies more as panels shrink. Scoring every compound on every subpanel, as they did, mixes the ranking with each definition's convention for compounds that have no active target on the subpanel (Supporting Information, Section S1). Under D1, such a compound receives no score. We therefore compared all measures on the same random subpanels, from 10 targets up to the full panel, with the D1 gate applied to all of them (Figure 6). Coverage, the share of compounds scored on the full panel that are still scored on the subpanel, is then the same for every measure. At 50 targets it is 88% on Davis, 71% on Klaeger, 77% on Anastassiadis and 99% on Metz. We write p^* for the smallest panel size tested at which the mean Spearman correlation with the full-panel ranking, over compounds scored on both, exceeds 0.90, a bar borrowed from the "excellent" range of intraclass-correlation guidelines.[25]

How quickly a measure recovers depends on how much its ranking rests on the few strongest targets (Figure 6, top). On the full panel, the S-score with $\tau = x_0$, the entropy with $\beta = x_0$, the Gini and the candidate all rank compounds by the share of targets with detected binding (Spearman ≥ 0.98 with the S-score at $\tau = x_0$ on every dataset). They reach p^* at 20 targets on Davis, 50 on Klaeger and 30 on Anastassiadis. The S-score at τ^* and the active count need 30, 80 and 50, and the entropy with $\beta = \tau^*$ 40, 100 and 65. The measures dominated by the strongest targets recover slowest. The published entropy needs 300 of 363 targets on Davis, 200 of 217 on Klaeger and 150 of 172 on Metz, and the ratio needs the full panel on every dataset except Klaeger (200). The published entropy gives binding below the activity threshold a small positive weight, so including weak binding does not by itself make a measure recover quickly. On Metz, where untested targets are set to the floor (Methods), the order of the first two groups reverses (65 against 50 targets). The candidate recovers

rankings no better than the S-score with $\tau = x_0$.

The active count and the S-score at τ^* give identical recovery on every dataset, although only the count satisfies D3. Recoverability is limited mainly by coverage. Keeping 90% of clearly separated pairs takes 150 targets on Davis, 200 on Klaeger and Anastassiadis, and 40 on Metz for every measure except the ratio and the published entropy, which need as many or more (Figure 6, middle). On Davis, Anastassiadis and Metz, compounds whose score changes most when a single target is left out tend to have larger rank errors at 50 targets (Spearman $r = 0.21$–0.73, for the entropy with $\beta = x_0$, the Gini and the candidate). On Klaeger this association is weak ($r \leq 0.14$).

Why the Smooth Hinge Matters

The entropy can violate D3. Its hard cutoff gives a kinase just above the baseline β a weight close to zero. A profile with many such kinases therefore has high entropy, and adding one kinase that is bound more strongly, but still below τ^*, concentrates the distribution and raises the selectivity score. For example, with $\beta = 5.0$ and $\tau^* = 6.0$, a profile with one kinase at $\mathrm{p}K_d = 7.0$ and fifty at 5.1 has an entropy-based selectivity of -4.89 bits, and adding one kinase at $\mathrm{p}K_d = 6.0$ raises it to -4.83. On Davis, Klaeger, and Metz, adding one kinase at τ^* never raises the entropy's score for a compound with an active kinase, at any baseline in their sweeps. On Anastassiadis it does at baselines below the 20% floor. With raw percent inhibition as weights ($\beta = 0\%$), it raises the score for 33% of the 133 compounds with an active kinase, and for 16% at $\beta = 5\%$, but for none at 15% or above. Adding real kinases rarely does this: expanding random 50- or 100-kinase panels by 1 to 30 kinases that a compound binds below threshold raises the entropy's score in at most 0.1% of cases. With $\beta = \tau^*$, the entropy cannot violate D3, because it gives no weight to binding below τ^*.

The candidate cannot violate D3. By the proposition in Methods, no sub-threshold addition can raise its score, and its condition on T is sharp: just below it, a profile with one strong target and enough kinases just above the floor violates D3. Below the bound,

constructed profiles of this kind violate D3 (by up to 0.014 bits on Anastassiadis with $T = 10$), although none of the real compounds does.

Discussion

Our characterization shows that the rankings given by kinase inhibitor selectivity definitions depend on analysis choices that are usually left implicit: which compounds are scored, how affinities are weighted, how pairs without detected binding are treated, where the activity boundary is placed, and which targets the panel includes. On a common log-affinity scale, the ratio and the distribution-based family (S-score, entropy, Gini) produce substantially different compound rankings, and the largest disagreements are concentrated in identifiable kinds of profiles. This has practical consequences: a compound that ratio-based analysis designates as selective may be designated as moderately promiscuous by entropy-based analysis, and vice versa, depending on the shape of its binding profile. Among compounds that clear the activity threshold, the two families disagree most on broad profiles (Type 3, Results).

Ratio measures follow classical pharmacology and analytical chemistry, where selectivity is pairwise: IUPAC defines a selectivity coefficient as the ratio of binding constants for an analyte and an interferent.[26] Kinome-wide profiling instead asks how concentrated a compound's binding is across hundreds of kinases, the question the entropy and Gini address.

Practical Recommendations

Pending a formal resolution, four practical recommendations follow from our findings. First, D1 (reliability threshold) should be adopted as a reporting standard. Compounds with no binding above the $\mathrm{p}K_d > 6$ activity threshold should not receive selectivity scores, or should have their scores explicitly flagged as unreliable. Of the 22 such Klaeger compounds (Type 1), 16 bind some kinase above the floor of $\mathrm{p}K_d = 5.0$, so only a gate at the activity threshold excludes them.

Second, the activity boundary should be fixed in the analysis protocol, before the data are examined, and selectivity assessments should report how rankings change when it is moved. No measure we examined satisfies D2. Rank standard deviation across the parameter sweep is a compound-specific summary of this sensitivity, and its value depends on the range and spacing of the sweep. It is not a guide to robustness across datasets: on the 46 compounds profiled in both Davis and Klaeger, it is not associated with how far a compound's rank moves between the two ($|r| \leq 0.18$).

Third, a comparison of selectivity across studies that use different definitions should follow a protocol fixed in advance, naming the definitions, parameter values and compounds compared. Compounds on which the definitions disagree should be reported, not removed. Within the distribution-based family, the S-score and our log-affinity entropy and Gini correlate at 0.89–0.999 among compounds with an active kinase on Davis and Klaeger; the published entropy does not (Results). They give similar rankings when parameters are chosen consistently and panels have similar, near-complete coverage. Without the D1 gate, the Gini diverges on smaller panels because of its convention for undefined scores (Supporting Information, Section S1). Ratio-based measures, in turn, should not be compared directly with distribution-based measures without acknowledging the conceptual distinction.

Fourth, a selectivity analysis should state its unit of analysis and, for small panels, how many compounds remain scorable. Counting mutant constructs or complex subunits as separate targets changes the rankings (Supporting Information, Section S6). On random 50-target panels, 12–29% of the compounds scored on the full Davis, Klaeger or Anastassiadis panel have no active target.

Limitations

The analysis covers four profiling datasets spanning three assay technologies and two readout types, but these still represent a fraction of the chemical and kinase space relevant to drug discovery. The assays report on different scales and detect different shares of active interactions

(Methods). The two-family structure (in our common log-affinity operationalization) is clear only on the two pK_d datasets, and on Klaeger only among compounds with an active kinase. On Anastassiadis the separation comes mainly from compounds with no active kinase, and on Metz it largely reflects coverage (Results). The two instability relations tested on all four datasets (ratio–gap and entropy–n_{active}) hold with the qualifications given in Results, and on Metz the entropy relation may also reflect coverage (Methods). Findings should be interpreted in light of these technological differences.

Our comparison is also restricted to four widely used, historically established definitions. Bosc et al.'s window score and ranking score[7] were not included. The ranking score is $R(k)$ at a shifted rank (Methods), and the window score correlates strongly with the ratio on the same four datasets (Spearman $r = 0.91$–0.96), so neither forms a third cluster. The CATDS score[10] was not reimplemented. It is defined on Kinobeads concentration–response curves, for a designated target at a chosen concentration, and only Klaeger provides such curves. A proxy computed from our baseline-subtracted affinity profiles correlates more strongly with the Gini than with $R(k)$ on all four datasets (Spearman 0.91–0.98, against 0.72 or less, Supporting Information Table S2). The proxy lacks the concentration dependence of the published score, so this does not show how the published score behaves.

A further axis of platform dependence that this analysis does not examine is cell-based versus cell-free profiling. Selectivity classifications for the same compounds can differ substantially between biochemical binding panels and cellular target-engagement assays. For example, several type II (DFG-out-binding) inhibitors show cellular interactions that cell-free assays miss.[27,28] Davis profiles recombinant constructs, and Klaeger's Kinobeads approach captures endogenous, full-length kinases from cell lysate, so neither dataset represents the intact cell.

We compared how much the choice of dataset and the choice of selectivity definition each move a compound's selectivity ranking. We resolved 46 compounds profiled in both Davis (competition binding) and Klaeger (chemoproteomics) by matching PubChem CIDs against

Klaeger's drug names, restricted both panels to the 200 kinases they share, and computed each definition's median rank on this fixed set of 46 compounds (separately on each dataset). For the same definition, the Davis and Klaeger rankings correlate at Spearman $r = 0.33$ to 0.84 (mean 0.60), against $r = -0.09$ to 1.00 between definitions on the same dataset (mean 0.56). On these compounds, the choice of definition therefore moves rankings about as much as the change of dataset. Among the 40 compounds active in both datasets, the means are 0.72 and 0.73, and the Gini's cross-dataset correlation rises from 0.45 to 0.84, so part of its disagreement between datasets comes from its convention for undefined scores. Without restricting the panels to their shared kinases, the cross-dataset correlations are lower (mean 0.53), so part of the difference between datasets comes from which kinases each profiles. The ratio is the extreme case: on the shared panel its Davis and Klaeger rankings for the same 46 compounds correlate at only $r = 0.33$ ($p = 0.02$). This should temper any cross-study comparison of selectivity rankings regardless of which definition is used.

All four existing definitions and the candidate measure introduced above treat a compound's affinity vector as an unordered multiset. Thus, two compounds with identical affinity distributions receive identical scores regardless of which specific kinases carry the off-target activity. A compound whose off-targets are pharmacologically inert and one whose off-targets are known safety-relevant kinases (e.g. a kinase implicated in a class toxicity) are therefore scored identically if their affinity distributions match. None of the measures discussed here incorporates target identity, and none accounts for exposure. A measured affinity alone does not establish whether a target is occupied in a patient, which also depends on the drug concentration achievable at a tolerated dose. A selectivity score is consequently, at best, a summary of *binding breadth*, and go/no-go decisions should weigh it alongside target-specific liability knowledge and exposure modeling.

We also examined whether in vitro selectivity is associated with clinical safety, correlating the selectivity rankings of the FDA-approved Klaeger compounds with the proportions of FAERS reports that were serious, fatal, or involved hospitalization ($n = 46$) and with FDA-

label discontinuation rates ($n = 33$). None was significantly associated with selectivity under any definition (all $|r| \leq 0.20$, all $p > 0.19$). The only nominal signal is a weak correlation between raw FAERS report counts and the ratio ranking ($r = 0.31$, $p = 0.04$), which may reflect prescription volume. FAERS proportions describe reports, not incidence among treated patients, so this analysis cannot isolate the effect of selectivity on toxicity.

Fixing the boundary at the floor x_0, as we do for every measure except the S-score and the active count, has a limit. In none of the four datasets is x_0 a measured detection limit. On Davis and Klaeger it is the value assigned to pairs with no detected binding, on Metz the lowest reported value (Methods), and on Anastassiadis a 20% inhibition noise level that we set. Fixing x_0 therefore removes the analyst's choice only to the extent that each dataset's encoding of non-detection is accepted as given.

The D2 sweeps cover only four datasets, and the recovery analysis uses only uniformly random subpanels. Panels chosen by kinase family, as many screening panels are, may recover rankings differently.

The three properties are also unlikely to be the complete set. A measure should be correctly oriented against a crude external proxy for the concept, such as the number of active kinases, since a measure can satisfy D1 and D3 and still rank promiscuous compounds as selective. A measure should declare its scale type and its unit of analysis, because both change rankings (Figure 1; Supporting Information, Section S6). Agreement across datasets profiling the same compounds can be tested, as the comparison above does.

Conclusion

We have characterized four kinase inhibitor selectivity definitions on four profiling datasets, each curated to one reference target per human protein kinase. Their rankings depend on choices usually left implicit: where the activity boundary is placed, which off-target the ratio references, how compounds with no binding above the activity threshold are scored, how

the entropy and Gini are operationalized, and which assay targets count as kinases. Rank instability is associated with binding profile shape, and among the most unstable compounds it has three structurally distinct sources. On a common log-affinity scale, the S-score, entropy and Gini agree more with one another than with the ratio, most clearly on the two pK_d datasets. Under the association-constant form of the original publication, the entropy does not cluster with the S-score and Gini, and how far it departs from them depends on how pairs without detected binding are treated.

We stated three properties for a selectivity measure: a reliability threshold, distributional consistency across placements of the activity boundary, and monotonicity under weak off-target addition. No measure we assess satisfies distributional consistency. Moving the activity boundary between the assay floor and the activity threshold reverses 8–13% of compound pairs under the S-score, entropy and Gini on three datasets, and over 20% on the fourth. The boundary must therefore be fixed in the analysis protocol and its effect reported. The reliability threshold is a reporting rule that any measure can adopt. Monotonicity separates the measures: it holds for the ratio and for simple measures such as a count of active kinases, but not for the S-score with a threshold above the floor, the Gini, or the entropy with a baseline below the activity threshold.

The candidate we propose, a gated, smooth-hinge entropy, satisfies the reliability threshold and monotonicity, which we prove, and weights binding below the activity threshold continuously. On these data it ranks compounds almost identically to the entropy and to the share of targets with detected binding, which is also monotone, so it is not an empirical improvement on either. Recovery from small panels is a separate objective. Measures that follow the share of targets with detected binding, the candidate among them, recover rankings from fewer targets than the ratio and the published entropy, which rest on the strongest targets. On small random panels, recovery is limited mainly by the share of compounds that can be scored.

The immediate practical implications are that selectivity assessments should state their

unit of analysis, fix the activity boundary in advance and report how rankings change with it, withhold scores from compounds with no binding above the activity threshold, and report how many compounds remain scorable on small panels. The longer-term implication is that the field needs a principled basis for choosing among selectivity definitions. Establishing properties that single out a measure, as Shannon's axioms do for entropy, is a natural next step, for which additional datasets may be required.

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Data and Software Availability

All data and software used in this study are publicly available. The Davis kinase inhibitor profiling dataset is available at `https://github.com/dingyan20/Davis-Dataset-for-DTA-Prediction`. The Klaeger et al. dataset is available as supplementary data to the original publication (Science, 2017, `https://doi.org/10.1126/science.aan4368`). The Anastassiadis et al. (*Nat. Biotechnol.* 2011) and Metz et al. (*Nat. Chem. Biol.* 2011) datasets are available as supplementary data to their original publications (`https://doi.org/10.1038/nbt.2017`;

https://doi.org/10.1038/nchembio.530). All analysis scripts, processed data files, and the LaTeX source for this manuscript are available at https://github.com/polinavino/kinase-selectivity-definitions; this repository includes the code and full per-definition results for the secondary clinical-outcome analysis (FAERS adverse-event report rates, obtained via openFDA queries, and FDA-label discontinuation rates, from FDA product labels, for the FDA-approved Klaeger compounds). No confidential data were used in this study. All results reported in this manuscript are fully reproducible from the publicly available data and code. A preprint of this manuscript is available on ChemRxiv at https://doi.org/10.26434/chemrxiv.15001618/v1.

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Conflict of Interest

The author declares no conflict of interest.

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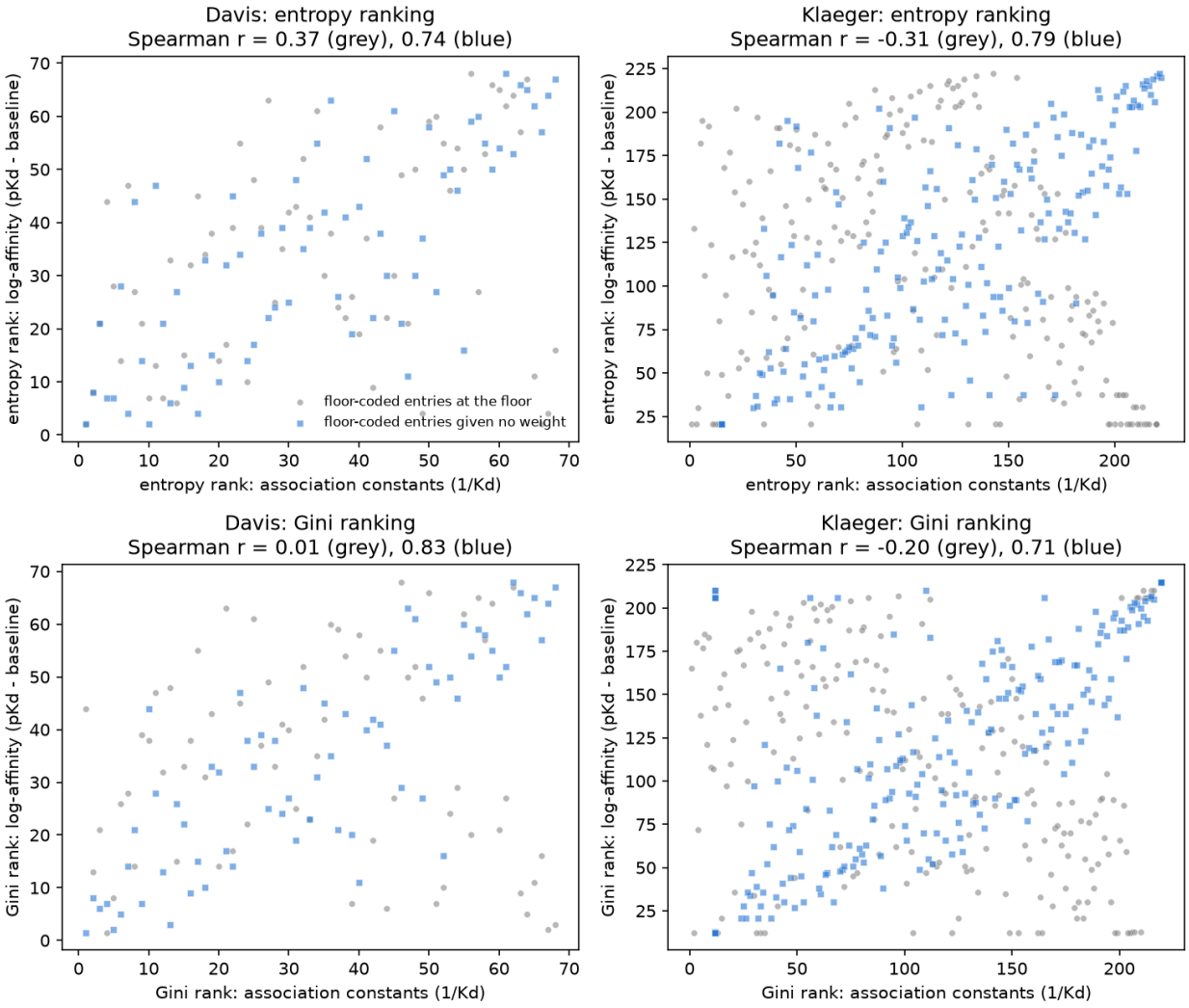


Figure 1: Entropy (top) and Gini coefficient (bottom) rankings on Davis and Klaeger under two operationalizations: association-constant weighting, the published form of the entropy (x-axis), and the baseline-subtracted log-affinity form used throughout (y-axis, median rank over the baseline sweep). Grey: floor-coded pairs kept at the floor; blue: floor-coded pairs given no weight. Spearman r above each panel.

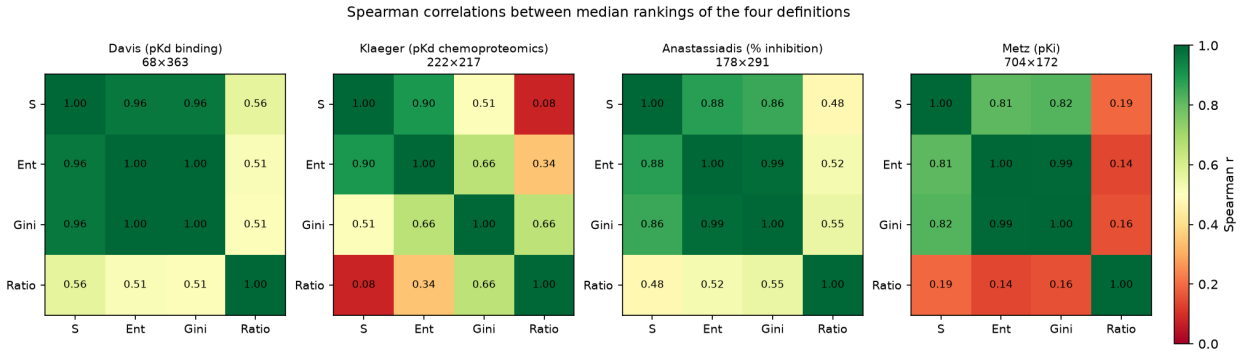


Figure 2: Spearman correlations between the median rankings of the four selectivity definitions on each dataset (compounds × targets above each panel). S: S-score; Ent: entropy. On Metz, rankings largely follow how many kinases each compound was measured against (Methods).

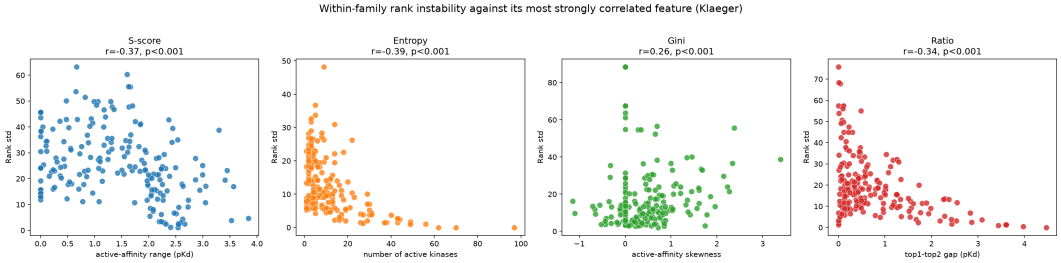


Figure 3: Within-family rank instability against the profile feature most strongly correlated with it, for each definition family (Klaeger, compounds with an active kinase, $n = 200$): active-affinity range (S-score), number of active kinases (entropy), active-affinity skewness (Gini), and top_1–top_2 gap (ratio). Spearman r above each panel.

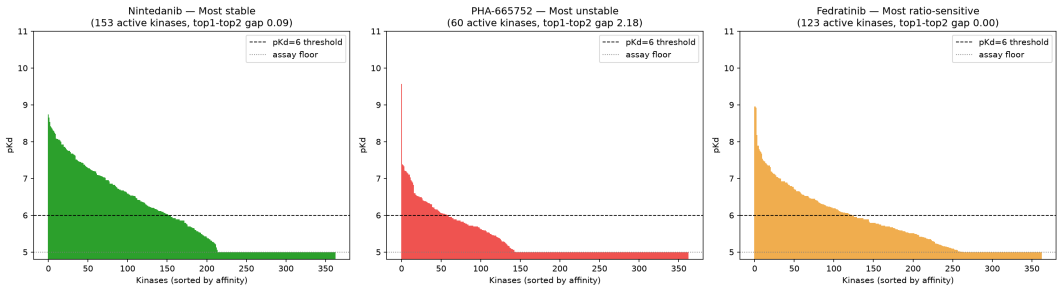


Figure 4: Binding profiles of three Davis compounds, targets sorted by affinity. Left: nintedanib, most stable ($\sigma = 2.1$), 153 active targets. Center: PHA-665752, most unstable ($\sigma = 16.4$), 60 active targets with one dominant target (Type 3). Right: fedratinib, most ratio-sensitive (Type 2): JAK2 and GAK tied, among 123 active targets. Dashed line: $\text{p}K_d = 6$ activity threshold; dotted line: $\text{p}K_d = 5$ assay floor.

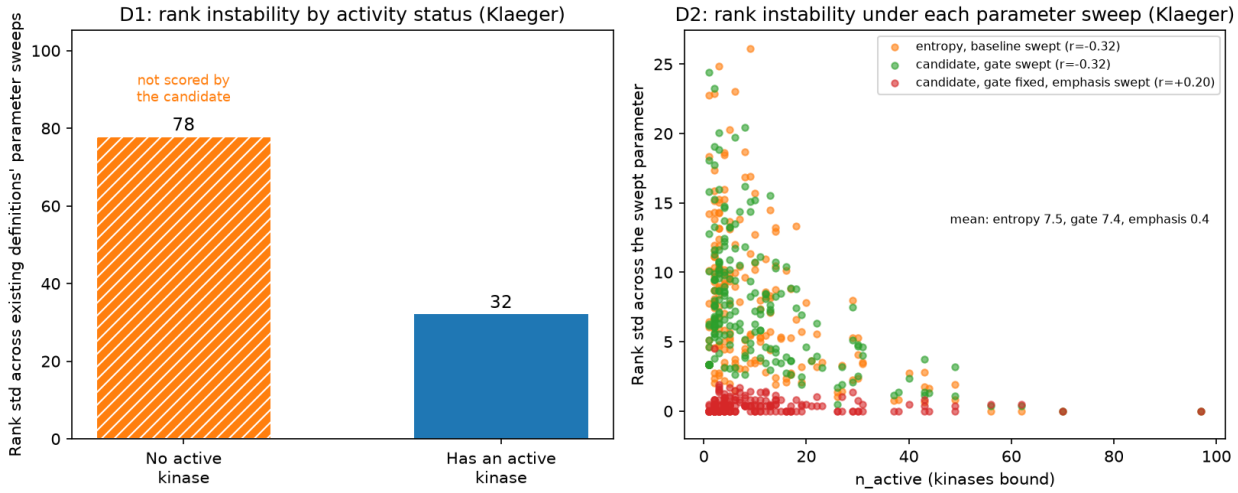


Figure 5: Rank instability on Klaeger. *Left (D1):* rank standard deviation across the existing definitions' parameter sweeps, for compounds with no binding above the activity threshold and for compounds with at least one active kinase. *Right (D2):* per-compound rank standard deviation against the number of active kinases, with one parameter swept over $B = [5.0, 6.0]$: the entropy's baseline β (orange), the candidate's gate x_0 with its emphasis baseline pinned to x_0 (green), and the candidate's emphasis baseline with x_0 fixed at the assay floor (red).

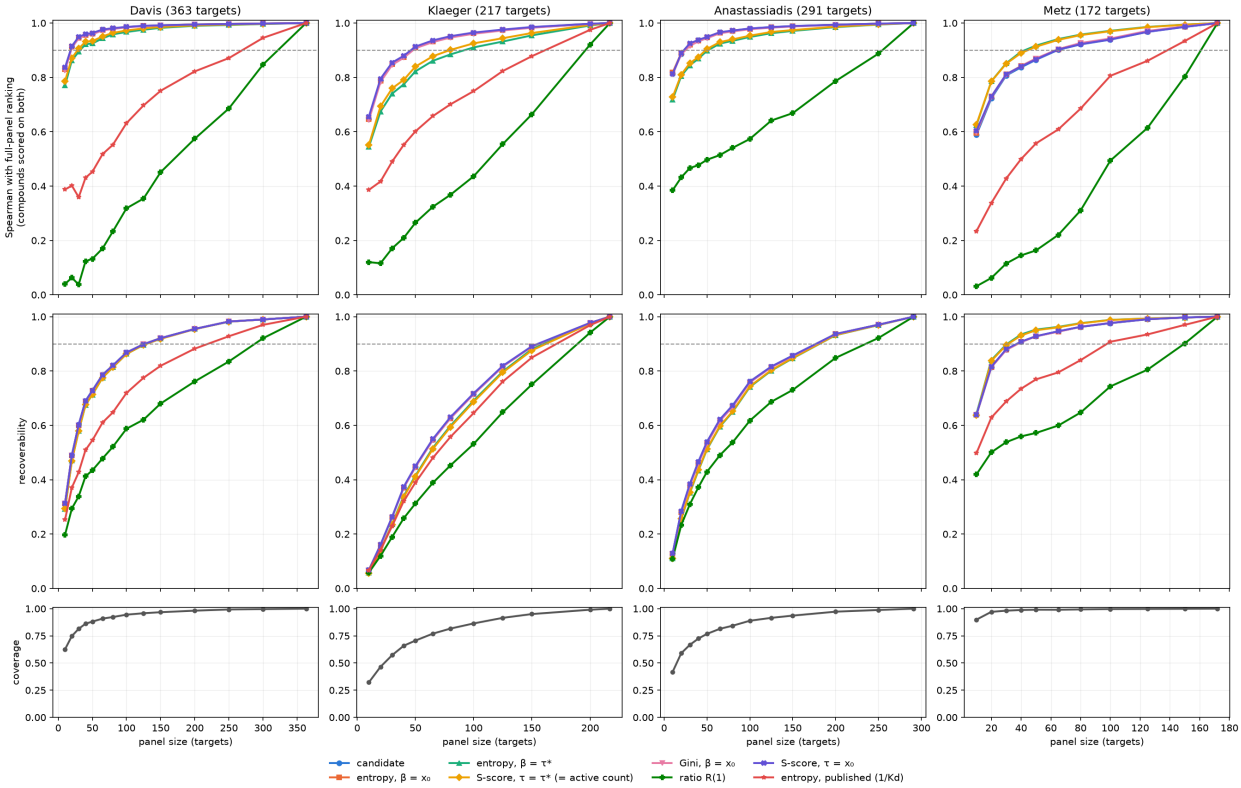


Figure 6: Recovery from random subpanels with the D1 gate applied to every measure, on all four datasets (50 subpanels per size). Top: mean Spearman correlation between the subpanel and full-panel rankings, over compounds scored on both. Middle: recoverability, the share of clearly separated full-panel pairs that are scored on the subpanel and keep their order. Bottom: coverage, the share of compounds scored on the full panel that are still scored. The S-score at τ^* and the active count coincide, and the S-score at x_0 is drawn over the candidate, which it almost matches. The published entropy is not defined on Anastassiadis. Dashed line: 0.90.